	CHAIN OF CUSTODY FORM (Original Stays With Samples)			US EPA Experimental Stream Facility 1003 US Highway 50, Milford, Ohio 45150 Phone # 513-965-4805 FAX # 513-965-4806					Originator:	
									Ship Date: 20180125	Contact:
									Destination: Cincinnati	Phone #:
									Recipient: Sam Smith	E-Mail:
									Phone #: 513 569 7681	
Collection Date YYYY-MM-DD	Collection Time HH:MM:SS	SiteID	LongID	CrossID	REP	Matrix Type*	TYPE (UKN, DUP, SPK, BLK)	Preservative	Analyses Required	
20180123	16:12:00				1	DI	FIELD BLK		TNO3, TNO2, TRP, TNH4, TN02, TP, TN	
20180124	09:53:00	FL	02	0.1	1	SW	UNK		"	
20180124	10:07:00	FL	05	0.1	1	SW	UNK		"	
20180124	10:40:00	FL	01	0.1	1	SW	UNK		"	
20180124	11:01:00	FL	07	0.1	1	SW	UNK		"	
20180124	11:50:00	FL	15	0.1	1	SW	UNK		"	
20180124	12:05:00	FL	14	0.1	1	SW	UNK		"	
20180124	12:20:00	FL	03	0.1	1	SW	UNK		"	
20180124	12:38:00	FL	04	0.1	1	SW	UNK		"	
20180124	12:53:00	FL	45	0.1	1	SW	UNK		"	
20180124	13:55:00	FL	43	0.1	1	SW	UNK		"	
20180124	13:09:00	FL	49	0.1	1	SW	UNK		"	
20180124	14:27:00	FL	47	0.1	1	SW	UNK		"	
20180125	12:50:00				1	DI	HOUSE BLK		"	
20180123	16:12:00				1	DI	FIELD BLK	2MHCl	TOC	
20180124	09:53:00	FL	02	0.1	1	SW	UNK	"	TOC	
20180124	10:07:00	FL	05	0.1	1	SW	UNK	"	TOC	
20180124	10:40:00	FL	01	0.1	1	SW	UNK	"	TOC	
20180124	11:01:00	FL	07	0.1	1	SW	UNK	"	TOC	
20180124	11:50:00	FL	15	0.1	1	SW	UNK	"	TOC	
20180124	12:05:00	FL	14	0.1	1	SW	UNK	"	TOC	
20180124	12:20:00	FL	03	0.1	1	SW	UNK	"	TOC	
20180124	12:38:00	FL	04	0.1	1	SW	UNK	"	TOC	
20180124	12:53:00	FL	45	0.1	1	SW	UNK	"	TOC	
20180124	13:55:00	FL	43	0.1	1	SW	UNK	"	TOC	
20180124	13:09:00	FL	49	0.1	1	SW	UNK	"	TOC	
20180124	14:27:00	FL	47	0.1	1	SW	UNK	"	TOC	
						~~DI~~				
	Relinquished By:		Date:	Time:	Received By:		Date:		Time:	

CHAIN OF CUSTODY FORM
(Original Stays With Samples)

US EPA Experimental Stream Facility
1003 US Highway 50, Milford, Ohio 45150
Phone # 513-965-4805
FAX # 513-965-4806

Originator:
Ship Date: 20180125
Destination: Cincinnati
Recipient: Sam Smith
Phone #: 513-569-7681

Collection Date YYYY-MM-DD	Collection Time HH:MM:SS	SiteID	LongID	Cross ID	REP	Matrix Type*	TYPE (UKN, DUP, SPK, BLK)	Analyses Required	*Volume of liquid sample filtered (mL)
20180124	09:53:00	FL	02	0.1	1	SW	UNK	Chlorophyll	281.66
20180124	10:07:00	FL	05	0.1	1	SW	UNK	"	345.7
20180124	10:40:00	FL	01	0.1	1	SW	UNK	"	277.82
20180124	11:01:00	FL	07	0.1	1	SW	UNK	"	244.68
20180124	11:50:00	FL	15	0.1	1	SW	UNK	"	252.79
20180124	12:05:00	FL	14	0.1	1	SW	UNK	"	228.14
20180124	12:20:00	FL	03	0.1	1	SW	UNK	"	189.56
20180124	12:38:00	FL	04	0.1	1	SW	UNK	"	185.73
20180124	12:53:00	FL	45	0.1	1	SW	UNK	"	187.91
20180124	13:55:00	FL	43	0.1	1	SW	UNK	"	116.03
20180124	13:09:00	FL	49	0.1	1	SW	UNK	"	136.67
20180124	14:27:00	FL	47	0.1	1	SW	UNK	"	142.74
20180123	16:12:00				1	DI	FIELD BLK	"	323.6
	Relinquished By:		Date:	Time:	Received By:				

GAS TRAP DATA SHEET

Lake:			11/24/18	11/25/18		
Site	**Lat**	**Long**	**Deployment time/date**	**Retrieval time/date**	**Volume recovered (mL)**	**Exetainer codes**
24	35.941921	78.588105	11/24 9:22	11/25 9:12	0	—
16	35.956731	78.590470	9:42	9:25	0	—
2	35.976160	78.588878	9:53	9:38	0	—
5	35.970128	78.621245	10:07	9:54	0	—
12	35.965656	78.633950	10:15	10:03	7	—
13	36.003645	78.646338	10:33	10:18	0	—
01	36.002063	78.634468	10:40	10:28	0	—
21	36.008806	78.628016	10:47	10:36	0	—
07	36.005533	78.667895	11:01	10:52	0	—
11	36.019358	78.617038	11:15	11:10	0	—
15	36.022805	78.714566	11:50	11:17	0	—
14	36.059343	78.718600	12:05	11:30	0	—
03	36.032053	78.736593	12:20	11:46	3	—
08	36.018941	78.748813	12:35	11:58	0	—
04	36.013315	78.759838	12:3	12:10	0	—
10	36.039981	78.755815	12:46	12:24	0	—
45	36.044135	78.758165	12:53	12:34	0	—
57	36.048430	78.752220	13:02	12:45	0	—
49	36.051323	78.756903	13:09	12:54	2	—
61	36.053565	78.759526	13:15	13:03	0	—
55	36.053218	78.761710	13:19	13:10	1	—
54	36.057810	78.765323	13:26	13:20	0	—
63	36.056033	78.770086	13:37	13:35	2	—
43	36.061573	78.773126	13:55	13:50	0	—
59	36.062838	78.771125	14:05	13:58	24	—
58	36.068600	78.775748	14:15	14:10	0	—
47	36.068506	78.768395	14:27	14:19	0	—
68	36.069328	78.766726	14:30	14:28	0	—
56	36.073070	78.766220	14:34	14:35	0	—
52	36.079160	78.760936	14:41	14:48	0	—

DISSOLVED GAS AND AIR DATA SHEET

Lake:	FALLS LAKE				Date: 1/25/18			
¥Site	Dissolved gas Exetainer codes		Water Vol/ Helium Vol	Water Temp (degC)	Atmospheric Air Exetainer Codes			BP (mm Hg)
2	01242018.FI.2.DG.1	DG.2	115/25	6.6	01242018.FI	AA.1	FI.2.AA.2	FI.2.AA.3
5	5.DG.1	5.DG.2		5.9				
1	1.DG.1	1.DG.2		5.7				
7	7.DG.1	7.DG.2		5.7				
15	15.DG.1	15.DG.2		5.7				
14	14.DG.1	14.DG.2		8.3				
3	3.DG.1	3.DG.2		7.9				
4	4.DG.1	4.DG.2		7.9				
45	45.DG.1	45.DG.2		9.7				
49	49.DG.1	49.DG.2		9.3				
43	43.DG.1	43.DG.2		11.1				
47	47.DG.1	47.DG.2		9.3	47.AA.1	47.AA.2	47.AA.3	47.AA.4
					↳ 2GR Ambient		14:33	3.5

FI.2.AA.4 FI.2.AA.5 47.AA.5

¥Site: all dissolved gas samples collected at 0.1m depth

Boat motor should be off for 2 minutes before sample collection

Water Vol/Air Vol :: 115ml/25ml

Goal is to inject 20mL of gas into exetainer

Signature Date

GREENHOUSE GAS ANALYZER AND GPS DATA SHEET

FILE NAMES	
Lake:	FALLS LAKE
Sample Date:	1/25/18
GGA file	394-2018-01-25_f0000.txt
gps file name	

Clock GPS 14:31
IPAD LGR 09:29:30

DEPLOYMENT DATA

Sample site	Chamber start time	Bubbling observed (Y/N)	GPS logger started	Chamber volume (L) - Level	Water Temperature (C)	CH_4 and CO_2 profile checked	
24	09:12:53	✓	✓	50	6.0	✓	
16	09:25:15	✓	✓	50	6.0	✓	
2	09:38:45	✓	✓	50	6.6	✓	
5	09:54:15	✓	✓	50	6.5	✓	
12	10:03:44	✓	✓	50	6.3	✓	
13	10:18:00	✓	✓	50	6.3	✓	
1	10:28:00	✓	✓	50	6.3	✓	
21	10:36:26	✓	✓	50	6.3	✓	Very Turbid
7	10:52:30	✓	✓	50	6.5	✓	
11	11:10:20	✓	✓	50	6.4	✓	
15	11:17:00	✓	✓	50	6.5	✓	
14	11:30:50	✓	✓	50	6.6	✓	
3	11:46:50	✓	✓	50	7.3	✓	
8	11:52:09	✓	✓	50	7.5	✓	
4	12:10:20	✓	✓	50	8.1	✓	
10	12:24:20	✓	✓	50	7.4	✓	
45	12:34:37	✓	✓	50	9.5	✓	
57	12:45:15	✓	✓	50	9.6	✓	
49	12:54:35	✓	✓	50	9.7	✓	
61	13:03:30	✓	✓	50	9.0	✓	
55	13:10:58	✓	✓	50	8.0	✓	
54	13:20:40	✓	✓	50	8.1	✓	Very Turbid
63	13:35:30	✓	✓	50	8.3	✓	
43	13:50:05	✓	✓	50	8.9	✓	
59	13:58:20	✓	✓	50	8.5	✓	Very Turbid
58	14:10:20	✓	✓	50	8.0	✓	
47	14:19:50	✓	✓	50	9.2	✓	
68	14:27:50	✓	✓	50	9.5	✓	
56	14:35:40	✓	✓	50	9.8	✓	
52	14:48:40	✓	✓	50	9.5	✓	

SONDE DATA SHEET

Lake: Falls Lake Sample Date: 1-24-2018

Site	Site Depth (m)	Sample Depth (m)	Temp (C)	DO (mg/l)	DO (%sat)	sp.Cond (µs cm^{-1})	pH	Chl a (ug/L)	Turbidity (NTU)	Notes
24	2.5	0.1	5.89	11.1	89.3	0.085	7.39	5.5	3.5	
16	8.0	0.1	6.06	11.92	95.9	0.085	7.2	6.9	3.9	
2	4.0	0.1	7.88	11.70	98.5	0.089	7.35	4.7	12.2	
5	6.0	0.1	6.14	11.5	92.5	0.088	7.4	5.1	3.5	
12	5.0	0.1	5.79	11.90	94.8	0.090	7.36	7.6	3.4	
13	6.5	0.1	6.38	12.62	102.5	0.092	7.61	9.6	3.2	
01	3.0	0.1	6.28	12.50	101.2	0.091	7.52	11.9	4.3	
21	4.7	0.1	8.40	11.84	101.0	0.083	7.58	3.3	91.0	Muddy water
07	7.7	0.5 ft	6.55	12.96	105.7	0.103	7.56	6.6	4.1	
07		3.3 ft	6.46	13.13	106.6	.103	7.55	12.2	4.0	
07		6.6 ft	6.44	13.15	106.9	.103	7.54	23.0	4.1	
07		9.9 ft	6.30	13.1	105.8	.103	7.54	22.6	4.0	
07		13.2 ft	6.15	12.9	104.2	.103	7.53	25.0	4.4	
07		16.5 ft	6.04	12.87	103.5	.103	7.52	24.7	5.0	
07		19.8 ft	5.98	12.66	101.8	.103	7.53	24.2	4.2	
07	8.0	22.6 ft	5.88	12.52	100.1	.103	7.51	24.0	25.0	
11	2.2	0.1m	6.59	13.0	106.1	.119	7.64	6.9	4.1	
15	3.5	0.1	6.67	12.95	105.6	0.125	7.57	5.8	4.4	
14	1.0	0.1	4.88	12.6	108.3	.112	7.6	6.4	3.9	
03	1.0	0.1	8.34	12.4	105.6	.162	7.62	5.0	4.2	
08	3.0	0.1	7.24	12.55	103.9	.151	7.69	9.3	6.0	
04	1.0	0.1	8.57	12.45	106.7	.148	7.75	8.2	12.3	
10	1.3	0.1	7.96	12.92	104.0	0.148	7.65	5.9	9.5	
45	1.5	0.1	7.98	12.80	108.4	0.206	7.69	9.8	9.2	
57	1.3	0.1	9.91	12.93	114.8	0.269	7.79	5.6	16.0	

49
55
61

SONDE DATA SHEET

Lake: Falls Lake. Sample Date: 11/24/18

Site	Site Depth (m)	Sample Depth (m)	Temp (C)	DO (mg/l)	DO (%sat)	sp.Cond (µs cm^{-1})	pH	Chl a (ug/L)	Turbidity (NTU)	Notes
49	1.8	0.1	9.94	12.12	107.5	0.314	7.83	9.2	28.7	
55	1.1	0.1	9.28	11.88	103.9	0.361	7.81	21.8	40.8	water l. muddy
61	1.5	0.1	9.74	11.80	104.0	0.321	7.8	7.5	36.0	water v.l.ty muddy
54	0.8m	0.1	9.19	11.69	101.8	0.374	7.75	9.3	59.5	-11-
63	1.4	0.1	10.21	12.3	109.8	0.360	7.77	4.6	32.7	-11-
43	1.0	0.1	9.12	11.44	99.2	0.386	7.82	13.2	77.4	-11-
59	1.5	0.1	9.98	10.61	93.5	0.283	7.82	10.5	153.7	-11-
58	1.0	0.1	9.52	11.21	97.7	0.241	7.73	8.7	120.0	-11-
47	1.1	0.1	11.09	12.17	110.7	0.313	7.76	11.8	35.9	
68	1.3	0.1	10.49	11.78	105.9	0.313	7.81	15.5	1 2	
56	1.5	0.1	11.09	12.29	111.7	0.268	7.92	16.6	35 1	
52	0.8	0.1	10.65	11.78	105.7	0.724	7.95	8.8	41.6	

Pre deployment CAL

Date 1/23/18
Cal cylinder serial # EPA # A20808
Dilution system serial # CC163816
LGR file 999-2018-01-23_f0000.txt

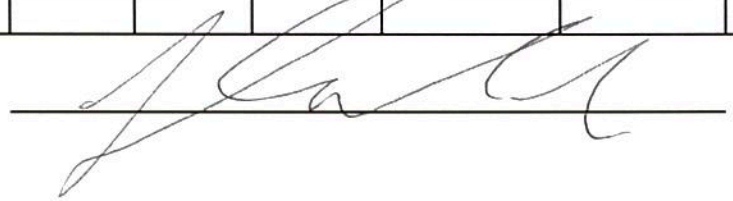
Zero Air Flow (Lpm)	Cal Gas Flow Readout (sccm)	CH4 cal (ppm)	CO2 cal (ppm)	CH4 resp (ppm)	CO2 resp (ppm)	Start Time	Stop Time
3.25	0.0	0.0	0.0	0.011	0.20	15:00	15:03:34
9	7.0	2.4	23.7	2.37	23.50	15:04:00	15:11:30
8.5	7.0	2.5	25.1	2.52	24.8	15:11:30	15:15:10
8	7.0	2.7	26.7	2.68	26.5	15:15:10	15:18:00
7.5	7.0	2.8	28.4	2.85	28.2	15:18:00	15:21:45
7	7.0	3.0	30.5	3.06	30.2	15:21:45	15:25:45
6.5	7.0	3.3	32.8	3.28	32.5	15:25:45	15:28:45
6	7.0	3.6	35.5	3.541	34.9	15:28:45	15:31:40
5.5	7.0	3.9	38.8	3.85	38.1	15:31:40	15:35:00
5	7.0	4.3	42.7	4.24	41.8	15:35:00	15:38:00

Signature _____

Post deployment CAL

Date 1/30/18
Cal cylinder serial # CC163816
Dilution system serial # EPA # A20808
LGR file 959-2018-01-30-f0000.txt

Zero Air Flow (Lpm)	Cal Gas Flow Readout (sccm)	CH4 cal (ppm)	CO2 cal (ppm)	CH4 resp (ppm)	CO2 resp (ppm)	Start Time	Stop Time
3.25	0.0	0.0	0.0	0.0135	0.02	16:33:30	16:35:45
9	7.0	2.4	23.7	2.40	23.8	16:38:00	16:43:25
8.5	7.0	2.5	25.1	2.53	25.0	16:43:25	16:46:00
8	7.0	2.7	26.7	2.68	26.5	16:46:00	16:48:30
7.5	7.0	2.8	28.4	2.85	28.4	16:48:30	16:50:45
7	7.0	3.0	30.5				
6.5	7.0	3.3	32.8				
6	7.0	3.6	35.5				
5.5	7.0	3.9	38.8				
5	7.0	4.3	42.7				

Signature 

ATTACHMENT A
YSI 6600 SONDE CALIBRATION LOG SHEET

SONDE ID: STD 16 **DATE:** 1/23/18 **TECHNICIAN:** JTW

DATE / TIME OK? (YES) / NO

BATTERY VOLTS: 12.14 (Change if < 10.0 v) **CHANGED BATTERIES?** YES / (NO)

TURBIDITY WIPER: PARKS 180° FROM OPTICS? (YES) / NO **CHANGED?** YES / (NO)

CHLOROPHYLL WIPER: PARKS 180° FROM OPTICS? (YES) / NO **CHANGED?** YES / (NO)

DISSOLVED OXYGEN: "DO Sat %" **BAROMETRIC PRESSURE:** 744.4 mm Hg

SENSOR OUTPUT TEST: PASS / FAIL (For Rapid-Pulse Probes, change membrane if failed)

Record first 10 readings from Discrete Sampling:

_____ _____ _____ _____ _____

_____ _____ _____ _____ _____

DO MEMBRANE CHANGED? YES / (NO) **PROBE RE-SURFACED?** YES / (NO) **DATE/TIME:** _____

[~97] **INITIAL:** 99.3 **CALIBRATED:** 98.0 **POST-DEPLOYMENT:** _____

? **DO CHARGE** [50 ± 25]: _____ **DO GAIN** [1.0 ± 0.35]: 1.05275

DEPTH: "Pressure" [m] (Ft) **PRESSURE OFFSET** [VENTED = 0 ± 6 / NON-VENTED = -14.7 ± 6]: 0.2567 (PSI)

[0] **INITIAL:** -0.056 **CALIBRATED:** 0.000 **POST-DEPLOYMENT:** _____

CONDUCTIVITY: "SpCond" [mS/cm] **CONDUCTIVITY CELL CONSTANT** [5.0 ± 0.45]: 4.88851 (Standard is 1.413 mS/cm @ 25°C)

1.413 ~~[10]~~ **INITIAL:** 1.~~413~~ 121 **CALIBRATED:** 1.413 @ 21.03 **POST-DEPLOYMENT:** _____

TURBIDITY: "Turb+NTU" [NTU]

[0] **INITIAL:** 0.0 **CALIBRATED:** 0.0 **POST-DEPLOYMENT:** _____

[~~1000~~] 126 **INITIAL:** 125.2 **CALIBRATED:** 126 **POST-DEPLOYMENT:** _____

CHLOROPHYLL: "Chlor ug/L" [ug/L]

[0] **INITIAL:** 0.0 **CALIBRATED:** 0.0 **POST-DEPLOYMENT:** _____

~~[113]~~ **INITIAL:** 6.7 **CALIBRATED:** _____ **POST-DEPLOYMENT:** _____

Site 2 Started 1/13/18

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pH: "pH"

[7] **INITIAL:** ~~6.83~~ 7.05 **CALIBRATED:** 7.0 **POST-DEPLOYMENT:** _____ **mV** [0 ± 40]: -38.8 @ 7.05

[10] **INITIAL:** 10.06 **CALIBRATED:** 10.0 **POST-DEPLOYMENT:** _____ **mV** [-177 ± 40]: -208.9 @ 10.06

ORP: "ORP mV" **ORP mV OFFSET [0 ± 100]:** _____